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Remaining useful life prediction of PEMFC based on matrix long short-term memory

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ABSTRACT

Prognostics and health management play a crucial role in ensuring the safety and maintenance of fuel cell vehicles. To address the insufficient prediction accuracy in existing models when handling long-term sequences and high-dimensional data, this paper proposes a proton exchange membrane fuel cell (PEMFC) remaining useful life prediction method based on a matrix long short-term memory (M-LSTM) network. Specifically, the Spearman correlation coefficient analysis is first employed to quantitatively evaluate the multidimensional correlations within the experimental data, identify key variables highly associated with fuel cell degradation, optimize feature selection, and achieve dimensionality reduction. Subsequently, an M-LSTM life prediction model is developed, incorporating an exponential gating mechanism and a matrix storage unit to enhance the global modeling capability of data-driven methods in complex nonlinear feature learning and long-sequence data processing. The results demonstrate that, compared with the LSTM model, under single-step prediction, M-LSTM reduces MAE and MSE by 76.3% and 58.2%, respectively, while increasing R^2 by 2.3%. Under multi-step prediction, M-LSTM reduces MAE and MSE by 79.4% and 60.4% respectively, while improving R^2 by 5.2%.

1. Introduction

In recent years, fuel cell vehicle technology using hydrogen as the main fuel is also considered to be one of the most important technical routes in the development of new energy vehicles [1–4]. Fuel cell vehicles have a short hydrogen refueling time and strong endurance. The only reactant in its operation is water, which has almost no pollution to the environment. Therefore, using fuel cell vehicles is a favorable way to maintain a clean environment and protect human health [5–7]. Although after years of research and continuous development, the proton exchange membrane fuel cell (PEMFC) installed in automobiles has made great research progress in terms of volume, low-temperature start-up, and energy efficiency, its durability problem has not yet reached the goal of industrialization and marketization [8,9]. Therefore, to accurately predict the degradation degree and remaining useful life (RUL) of PEMFC, extend the service life of fuel cells, and expand the application of fuel cell systems, prognostics, and health management methods have received widespread attention [10,11]. Currently, prediction methods are mainly divided into three categories: model-driven methods, data-driven methods, and hybrid models [12–14].

Model-based prediction methods can be divided into the use of degradation mechanism models, empirical models, or semi-empirical models [15]. Futter et al. [16] simulated the degradation process of the proton exchange membrane by introducing various chemical mechanisms in the battery, and verified the accuracy of the degradation model through an accelerated stress test (AST). This model belongs to a type of mechanism model. The introduction of chemical mechanisms greatly increases the complexity of modeling. Ou et al. [17] proposed a semi-empirical model-based prognostic method for predicting the degradation and remaining useful life of PEMFCs. The approach integrates a voltage prediction model, aging parameter analysis, and performance recovery after shutdown, validated through experimental cases. Although the semi-empirical model can reduce the complexity of modeling, the empirical data in it needs to be accumulated through experiments. Zhou et al. [18] proposed a real-time prediction (RTP)

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method for fuel cell lifespan by segmenting output power into distinct operating conditions, estimating voltage decay rates for each, and constructing an empirical model. Hybrid models rely on physical decay models and are therefore classified here as a special type of model-based approach. Pan et al. [19] introduced a novel hybrid technique combining an adaptive Kalman filter and a data-driven NARX neural network, where the Kalman filter and empirical aging model capture macroscopic degradation trends, while the NARX neural network provides fine-grained details. Liu et al. [20] designed an automatic machine learning algorithm integrating an evolutionary approach with an adaptive neuro-fuzzy inference system, enabling long-term degradation trend prediction under variable current loads and remaining life estimation. Wang et al. [21] developed a hybrid model that refines the input of a data-driven model using semi-empirical outputs, significantly reducing oscillations in long-term degradation predictions. However, fuel cell systems are complex physical and multi-scale systems. It is a difficult task to establish an accurate degradation model. In addition, multi-dimensional models require a lot of computing resources and time, which makes it difficult to apply to dynamic control and online diagnosis of fuel cells [22].

Therefore, many scholars have proposed data-driven prediction methods for fuel cell systems. Data-driven prediction methods refer to learning the degradation behavior of fuel cells from fuel cell aging experimental data through statistics or machine learning and using these learning results to predict the life of fuel cells [23]. Compared with model-based prediction methods, data-driven prediction methods can achieve high prediction accuracy without analyzing the degradation mechanism inside the fuel cell. LSTM is a special type of RNN (Recurrent Neural Network) designed to solve the shortcomings of RNN in dealing with long-term dependency problems. [24–26], which can avoid the problems of gradient explosion and vanishing. The fuel cell prediction method based on the LSTM architecture is currently a relatively effective means. Liu et al. [27] proposed a new method for predicting the remaining useful life (RUL) of PEMFC, which used an LSTM-based recurrent neural network combined with data smoothing and reconstruction techniques to improve prediction accuracy and reliability. Jia et al. [28] proposed a novel data-driven prediction model that integrates the multi-head self-attention (MHSA) mechanism and bidirectional long short-term memory.

However, LSTM also has some problems. First, the original intention of LSTM design is to overcome the long-term dependency problem of traditional RNN, but when processing very long sequences, LSTM may still perform poorly in capturing very long-distance dependencies. In particular, when the length of the sequence data exceeds a certain range, its performance will drop significantly. Second, LSTM needs to process sequence data time step by time, which makes it difficult to achieve efficient parallel computing. Compared with models such as Transformer, LSTM has weaker parallel computing capabilities on large-scale data, which limits its application in certain high-dimensional, large-scale tasks. In recent studies, Beck et al. [30] proposed a Matrix of Long Short-Term Memory (M-LSTM). Exponential gating is used in M-LSTM to efficiently modify stored values when more relevant information is encountered, overcoming a significant limitation of traditional LSTM. Matrix memory provides greater storage capacity, enabling M-LSTM to more efficiently handle rare labels, long-range dependencies, and complex data patterns. Finally, M-LSTM is fully parallelizable, enabling it to compute efficiently on modern hardware accelerators and scale to larger models. Table 1 is a comparison of the advantages and disadvantages of various life prediction methods.

Considering that fuel cell life data has the characteristics of high data dimension and high complexity, the computational requirements of neural networks are very high. Therefore, this paper studies a fuel cell life prediction method based on M-LSTM to alleviate the problem of large prediction result errors caused by the high dimension and high complexity of fuel cell life prediction data. Thanks to the exponential gating, matrix characteristics and parallelization of M-LSTM, the accuracy of fuel cell life prediction is improved.

Table 1
Comparison of life prediction methods for PEMFC.

Method	Advantages	Disadvantages
RNN	Simple structure	Unable to capture long-distance dependencies
LSTM	Solving the vanishing gradient problem	Requires a lot of data and time to train
Bi-LSTM	Capturing bidirectional dependencies	Requires a lot of data and time to train
LSTM-transformer [29]	Improved long-distance dependency modeling capabilities	High model complexity
M-LSTM	High-dimensional data processing capturing long-range dependencies	High computational complexity

The structure of the paper is as follows: Section 2 is the PEMFC aging experiment and the data processing. Section 3 is the M-LSTM prediction model in detail. Section 4 is the prediction results. Section 5 summarizes some important conclusions. The overall flow chart of the study is shown in Fig. 1.

2. Experiments and data processing

The fuel cell system is mainly composed of an air supply system, a hydrogen supply system, a water circulation system, and a high-voltage circuit, as shown in Fig. 2. The fuel cell platform used in this experiment has a rated power of 90 kW.

The stack consists of 400 cells connected in series. The experimental environment was set to standard atmospheric pressure, 25 °C, and 50% humidity. When measuring the polarization curve, the operating voltage range of the fuel cell stack was 0.46 V–0.75 V, and the current density varied from 0 A/cm^2 to 1 A/cm^2.The following provides an overview of each component:① Gas Supply System: The gas supply system ensures a continuous supply of reactant gases during the operation of the PEMFC stack. In this experiment, it is divided into a hydrogen supply unit and an air supply unit. Hydrogen is delivered from storage tanks, while air is compressed, filtered, and stored before being provided. ② Cooling System: The cooling system maintains the PEMFC stack's operating temperature within the optimal range, typically 30 °C–80 °C. At the start of the operation, when the stack temperature is low, the water tank automatically heats deionized water, which is circulated through the stack by a water pump to raise the temperature. Conversely, when the stack temperature becomes too high, the circulating water is cooled via cooling pipes and pumps to stabilize the system. ③ Electronic Load: To evaluate the performance and degradation of the PEMFC stack under real-world conditions, external electronic loads with higher current, voltage, and power ratings than the stack are employed. These loads simulate actual operating conditions and meet the requirements of the experimental design.

The fuel cell life data acquisition process is shown in Fig. 3. The life test conditions include four conditions: idle, start-stop, rated, and variable load. The experiment was carried out on a fuel cell bench, and a total of 1374 h of data was acquired. By plotting the polarization curve of the fuel cell, we can see that at the same current density, the fuel cell voltage begins to decrease over time, which shows that voltage is suitable as a parameter for characterizing fuel cell aging. In the following, the aging of the fuel cell will be evaluated by the decline of the fuel cell voltage. Considering that the current density of the fuel cell is 0.7 A/cm^2 when it is working, the voltage aging data when the current density is 0.7 A/cm^2 in the polarization curve is selected as the voltage aging data set.

The presence of considerable noise and irregular fluctuations in the experimental data set poses challenges to computational efficiency and accuracy. To address this, a wavelet thresholding technique is employed for data preprocessing. This method leverages the principle that,

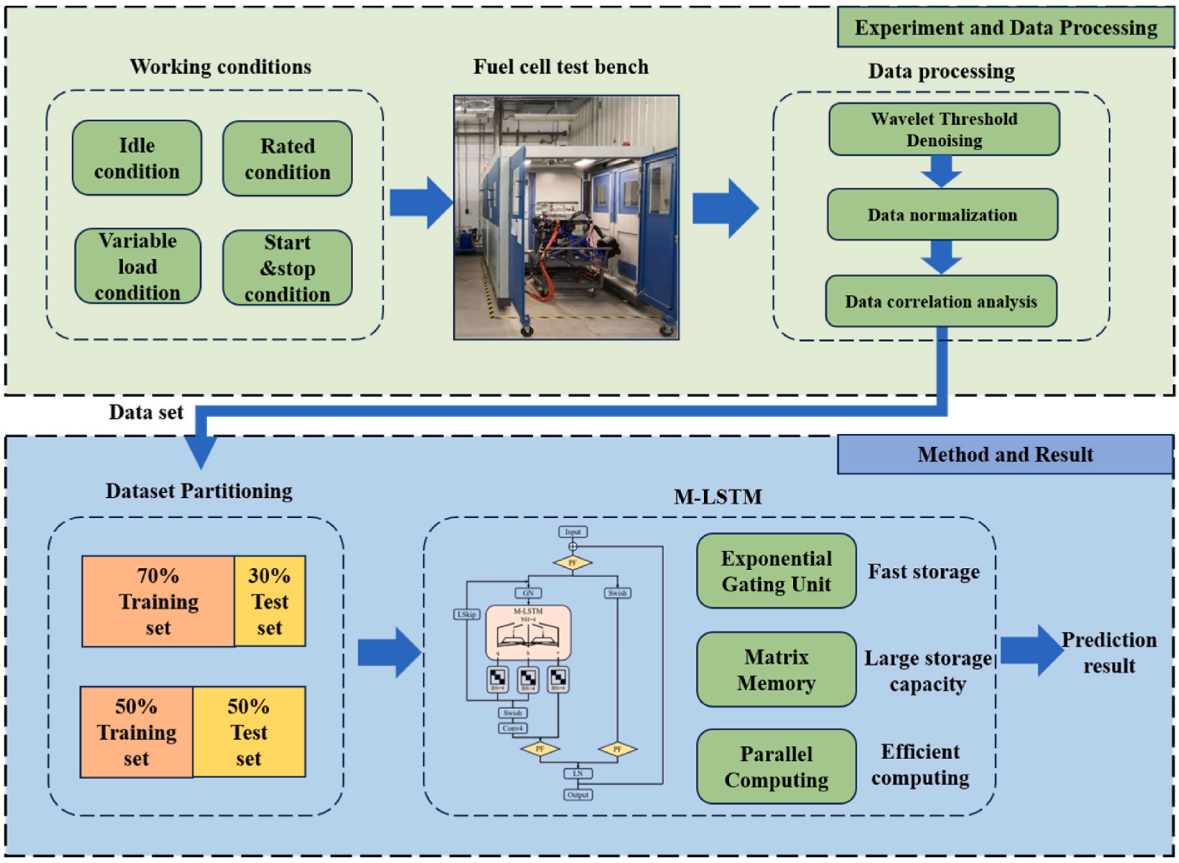


Fig. 1. Overall flow chart.

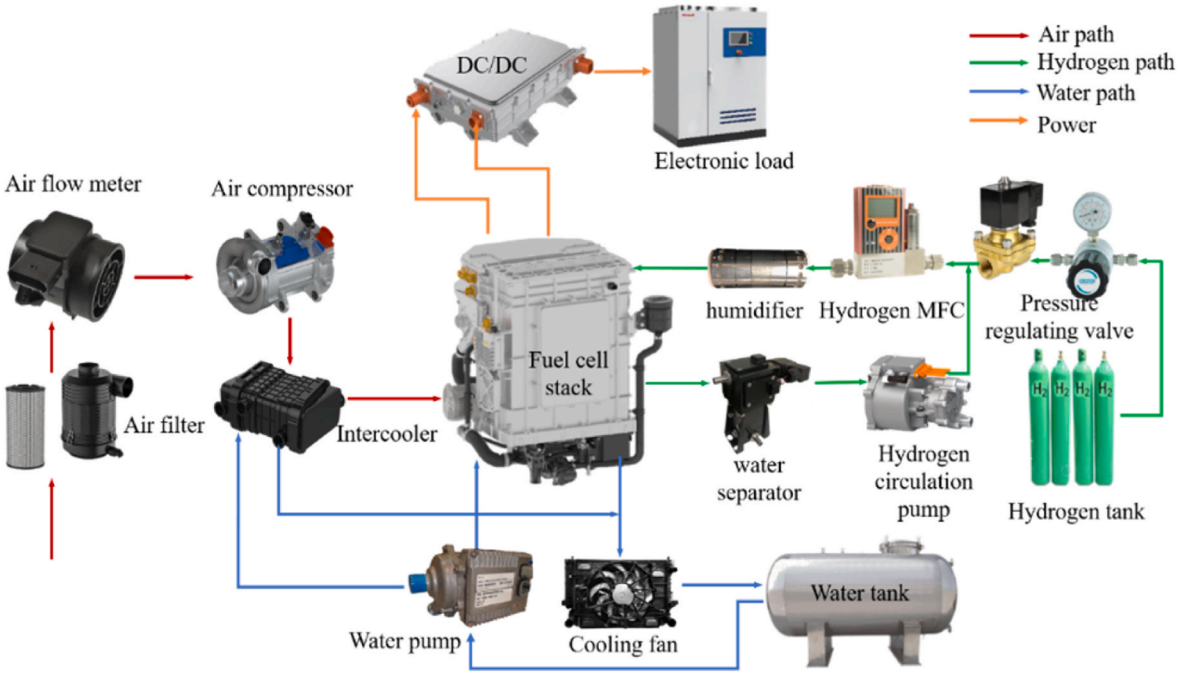


Fig. 2. Fuel cell system.

following multi-scale wavelet decomposition, the amplitude of signal wavelet coefficients typically surpasses that of noise. By establishing a predefined threshold, coefficients with amplitudes below this value are suppressed, while those exceeding the threshold are retained. Setting the wavelet base parameter to db5, the decomposition level to 8, the denoising method to minimax, the threshold rule to the soft threshold, and the threshold of each level to the default value. As depicted in Fig. 4, the denoising process effectively attenuates noise interference while

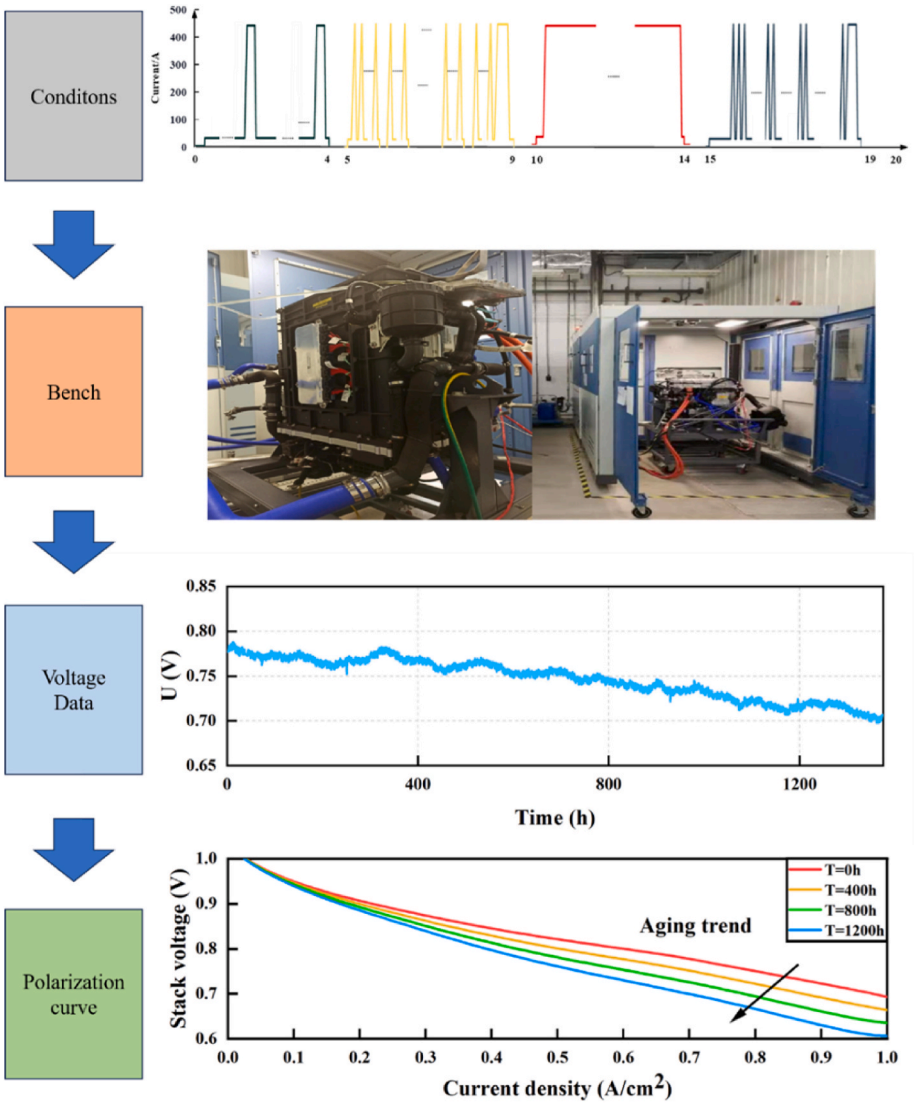


Fig. 3. Experimental data acquisition.

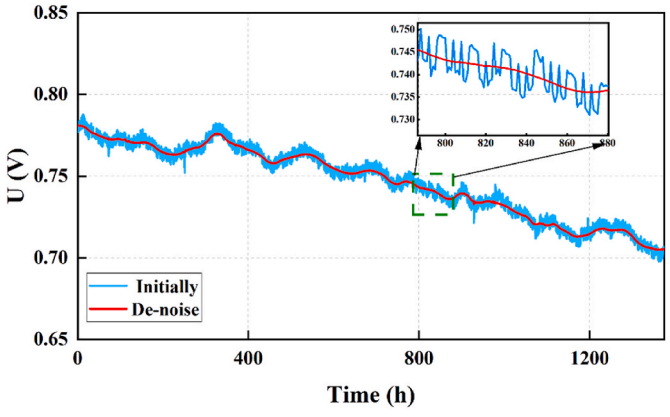


Fig. 4. Comparison of voltage waveforms before and after denoising.

preserving the critical degradation characteristics of the PEMFC stack, ensuring data integrity for subsequent analysis.

The high dimensionality of the data set (Table 2) introduces risks of distortion, complicating the accurate depiction of variable relationships. To mitigate these effects, this study applies a normalization process to the smoothed data, rescaling it within the [0,1] range. This approach reduces the influence of dimensional inconsistencies, enhancing the fidelity of variable correlation analysis.

Table 2
The aging parameters gathered during experiments.

Parameters	Physical meanings
U	The voltages of the stack(V)
J_i; I	The current density (A/cm^2) and current (A)
$T_{out_}$ H_2; $T_{in_}$ H_2	The temperatures of H_2 at outlet and inlet (°C)
$T_{out_}$ air; $T_{in_}$ air	The temperatures of air at the outlet and inlet (°C)
$T_{out_}$ water; $T_{in_}$ water	The temperatures of water at the outlet and inlet (°C)
$P_{out_}$ H_2; $P_{in_}$ H_2	The pressures of H2 at the outlet and inlet (Pa)
$P_{out_}$ air; $P_{in_}$ air	The pressures of air at the outlet and inlet (Pa)
$F_{out_}$ H_2; $F_{in_}$ H_2	The flow rates of H2 at outlet and inlet (L/min)
$F_{out_}$ air; $F_{in_}$ air	The flow rates of air at the outlet and inlet (L/min)
Air_ hyg	Air hygrometry at the inlet (%)
F_ water	The flow rate of cooling water (L/min)

As shown in Table 1, the PEMFC data includes a lot of parameters, including temperature, pressure, flow, etc. However, in the process of building the prediction model, too many parameters will lead to redundancy of the prediction model and affect the prediction speed of the model. Therefore, it is necessary to analyze and screen the data

obtained from the experiment and select the data most relevant to the voltage to balance the calculation speed and accuracy of the model. The Spearman rank correlation coefficient is widely recognized for its effectiveness in assessing relationships between time-series variables, especially when nonlinearity is involved. This study utilizes this method to evaluate the interdependence of data variables. The corresponding equation is as follows:

$$\rho = 1 - \frac{6\sum_{i=1}^{N} d_i^2}{N(N^2-1)} \tag{1}$$

Where, ρ is the rank correlation coefficient; d_i represents the element in the rank difference set, where $d_i = x_i - y_i$, $1 \le i \le N$.

Fig. 5 shows the correlation coefficient values between data variables. The correlation of data variables is positively correlated with the absolute value of their correlation coefficient. The absolute value of the correlation coefficient between fuel cell voltage and $T_{out_}H_2$, $T_{in_}H_2$, $T_{out_}$ air, $T_{in_}$ air, $T_{out_}$ water, $T_{in_}$ water, $P_{in_}$ air, $P_{out_}H_2$, $P_{in_}H_2$, $F_{in_}$ air, F_water less than 0.2, and it can be considered that these data have weak correlation. Therefore, these weakly correlated parameters are excluded to reduce the dimension of the data volume and improve the running speed of the model.

3. M-LSTM prediction model

3.1. M-LSTM

The storage capacity of LSTM is enhanced by replacing the scalar storage unit $c \in R$ with a matrix representation $c \in R^{d\times d}$, allowing retrieval operations through matrix multiplication. Moreover, a covariance update strategy is embedded within the LSTM design, where the forget gate modulates the decay factor, the input gate adjusts the learning rate, and the output gate modifies the amplitude of the retrieved output [30]. The M-LSTM forward pass is:

$$C_t = f_t \cdot c_{t-1} + i_t \cdot v_t \cdot k_t^{\mathrm{T}} \tag{2}$$

$$n_t = f_t \cdot n_{t-1} + i_t \cdot k_t \tag{3}$$

$$h_t = o_t \odot \widetilde{h}_t, \widetilde{h}_t = C_t \cdot q_t / \max\{|n_t^{\mathrm{T}} \quad q_t|, 1\} \tag{4}$$

$$q_t = W_q \cdot x_t + b_q \tag{5}$$

$$k_t = \frac{1}{\sqrt{d}} W_k \cdot x_t + b_k \tag{6}$$

$$v_t = W_v \cdot x_t + b_v \tag{7}$$

$$i_t = \exp(\widetilde{i}_t), \widetilde{i}_t = \omega_i^{\mathrm{T}} \cdot x_t + b_i \tag{8}$$

$$f_t = \sigma(\widetilde{f}_t) \; OR \quad \exp(\widetilde{f}_t), \widetilde{f}_t = \omega_f^{\mathrm{T}} \cdot x_t + b_f \tag{9}$$

$$o_t = \sigma(\widetilde{o}_t), \widetilde{o}_t = \omega_o \cdot x_t + b_o \tag{10}$$

Fig. 6 is a schematic diagram of the M-LSTM structure. The input is first projected upward through a projection factor for externalizing the output gate, followed by a causal convolution, and finally used as the input of the M-LSTM unit. The inputs q and k are obtained through a block diagonal projection matrix with a block size of 4, while the value v is directly input, skipping the convolution part. After mixing the M-LSTM sequence, the output is normalized by GroupNorm. Finally, a learnable skip input is added and the component is gated by an external output gate.

The specific process of PEMFC remaining service life prediction is as follows.

1) Take a group of points every 1 h to reconstruct the data set;
2) Use wavelet threshold denoising method to preprocess voltage data;
3) Use data normalization to process sample data;
4) Conduct correlation analysis on experimental data, exclude parameters with low correlation with voltage decay, and reduce the data latitude of the prediction model;
5) For the first prediction, the first 70% of the data set is selected as the training set, and the last 30% is selected as the prediction set;

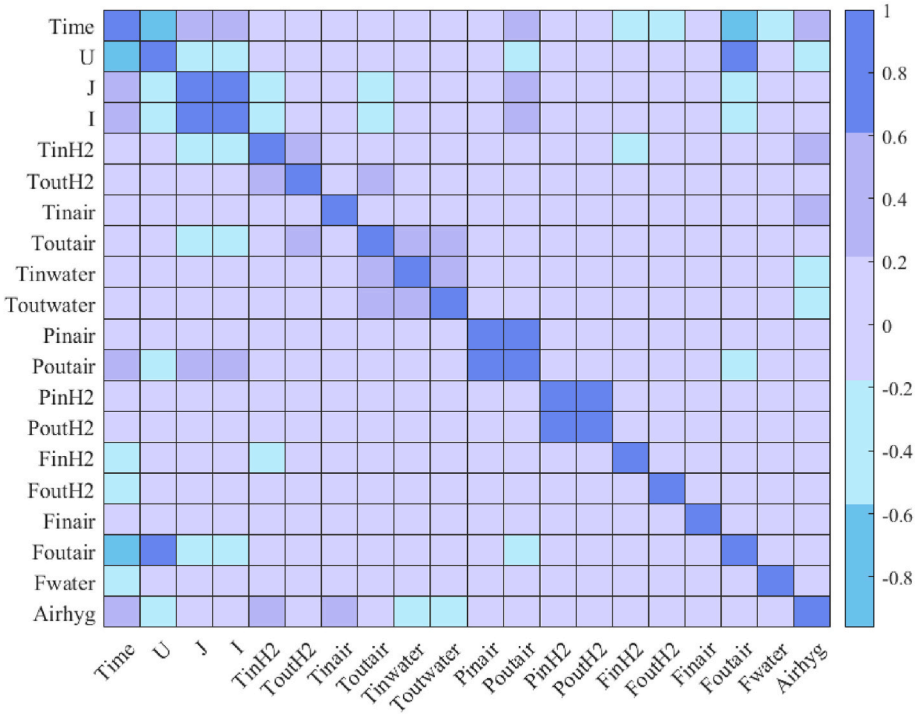


Fig. 5. Data filtering graph based on Spearman rank correlation coefficient value.

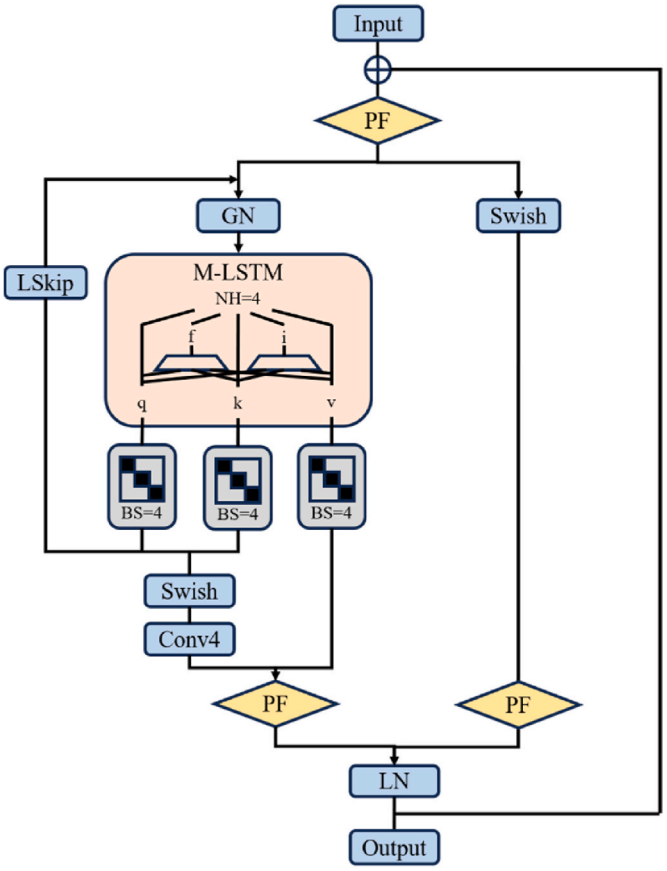


Fig. 6. Schematic diagram of M-LSTM structure.

6) For the second prediction, the first 50% of the data set is selected as the training set, and the last 30% is selected as the prediction set;
7) Establish LSTM, LSTM-Transformer, S-LSTM, M-LSTM model. Set the number of hidden layers, loss function, and optimizer;
8) Input the training set into models and set the number of training cycles;
9) The test set is input into the trained model, and single-step prediction and multi-step prediction are performed respectively. The step length of multi-step prediction is set to 20. The predicted value is obtained, compared with the actual value, and the prediction error is calculated.

3.2. LSTM

LSTM is a special RNN designed to solve the long-term dependency problem faced by traditional RNN when processing long sequence data. LSTM effectively saves and updates long-term information by introducing cell state and gating mechanism, thus solving the problem of gradient vanishing and gradient exploding. The structure of LSTM is shown in Fig. 7, which consists of input gate, forget gate and output gate. The specific formula is as follows:

$$C_t = f_t \cdot c_{t-1} + i_t \cdot z_t \tag{11}$$

$$h_t = o_t \cdot \widetilde{h}_t, h_t = \widetilde{\psi}(c_t) \tag{12}$$

$$z_t = \varphi(\widetilde{z}_t), \widetilde{z}_t = \omega_z^{\mathrm{T}} \cdot x_t + r_z \cdot h_{t-1} + b_z \tag{13}$$

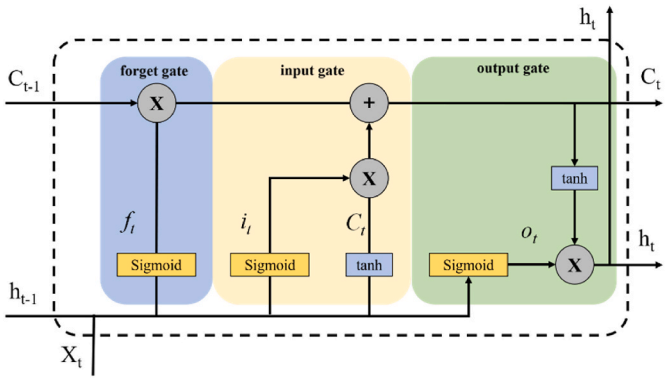


Fig. 7. Schematic diagram of LSTM structure.

$$i_t = \sigma(\widetilde{i}_t), \widetilde{i}_t = \omega_i^{\mathrm{T}} \cdot x_t + r_i \cdot h_{t-1} + b_i \tag{14}$$

$$f_t = \sigma(\widetilde{f}_t), \widetilde{f}_t = \omega_f^{\mathrm{T}} \cdot x_t + r_f \cdot h_{t-1} + b_f \tag{15}$$

$$o_t = \sigma(\widetilde{o}_t), \widetilde{o}_t = \omega_o^{\mathrm{T}} \cdot x_t + r_o \cdot h_{t-1} + b_o \tag{16}$$

Where, the weight vectors w_z, w_i, w_f, and w_o represent the input weights associated with the connections between the input x_t and the unit input, input gate, forget gate, and output gate, respectively. Similarly, r_z, r_i, r_f, and r_o denote the recurrent weights linking the hidden state h_{t-1} to the same components. The terms b_z, b_i, b_f, and b_o are the corresponding biases, while φ and ψ refer to the activation functions for the unit input and hidden state.

3.3. S-LSTM

To give LSTM the ability to modify storage decisions, introduce exponential gating, namely "exp" in equations (12) and (13). In particular, the input gate and forget gate can have exponential activation functions [30]. The forward pass of S-LSTM is:

$$C_t = f_t \cdot c_{t-1} + i_t \cdot z_t \tag{17}$$

$$n_t = f_t \cdot n_{t-1} + i_t \tag{18}$$

$$h_t = o_t \widetilde{h}_t, \widetilde{h}_t = c_t / n_t \tag{19}$$

$$z_t = \varphi(\widetilde{z}_t), \widetilde{z}_t = \omega_z^{\mathrm{T}} \cdot x_t + r_z \cdot h_{t-1} + b_z \tag{20}$$

$$i_t = \exp(\widetilde{i}_t), \widetilde{i}_t = \omega_i^{\mathrm{T}} \cdot x_t + r_i \cdot h_{t-1} + b_i \tag{21}$$

$$f_t = \sigma(\widetilde{f}_t) \; OR \quad \exp(\widetilde{f}_t), \widetilde{f}_t = \omega_f^{\mathrm{T}} \cdot x_t + r_f \cdot h_{t-1} + b_f \tag{22}$$

$$o_t = \sigma(\widetilde{o}_t), \widetilde{o}_t = \omega_o^{T} \cdot x_t + r_0 \cdot h_{t-1} + b_o \tag{23}$$

Fig. 8 illustrates the architecture of the S-LSTM model. The input undergoes causal convolution with a projection factor, incorporating Swish activations for the input and forget gates. It is then processed through a block-diagonal linear layer comprising four distinct "heads," corresponding to the input, forget, and output gates, as well as unit updates. Finally, the output is refined via a gated MLP employing a GeLU activation function and a projection factor for parameter alignment.

4. Result and discussion

To assess the predictive performance of the developed machine

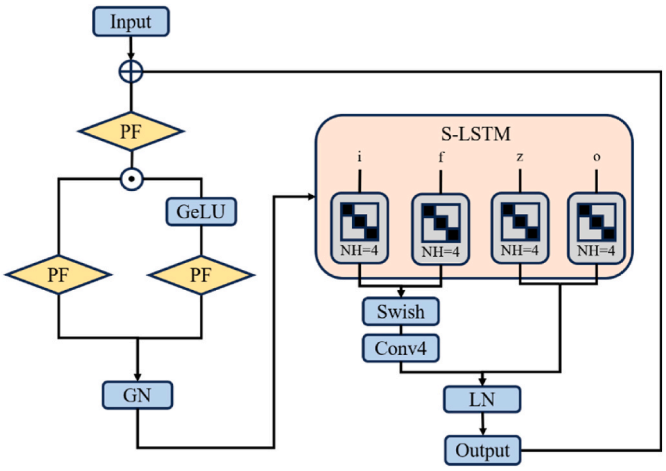


Fig. 8. Schematic diagram of S-LSTM structure.

learning model, this study employs Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2) metrics. Lower MAE and MSE values indicate higher prediction accuracy, while R^2values approaching 1 signify improved model reliability. The formulas for these metrics are as follows:

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|\hat{y}_i - y_i| \tag{24}$$

$$MSE = \frac{1}{n}\sum_{t=1}^{n}(\hat{y}_i - y_i)^2 \tag{25}$$

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (\bar{y}_i - y_i)^2} \tag{26}$$

where, $\hat{y}_i$ is the model output value, $\bar{y}_i$ is the average value, y_i is the true value.

Firstly, this paper uses 70% of the dataset as a training set and 30% as the test set. The prediction performance of LSTM, LSTM-Transformer, S-LSTM, and M-LSTM is compared. The prediction results of fuel cell degradation are shown in Fig. 9. The prediction results of S-LSTM and M-LSTM are closely related to the actual monitored voltage, and show higher prediction accuracy compared with LSTM-Transformer. In contrast, LSTM has the worst performance.

Fig. 10 shows the one-step forecast performance metrics for various forecast models using 70% of the training set. Among these, M-LSTM showed the best prediction performance, with a MAE of 0.5076×10^{-3}, an MSE of 0.1464×10^{-5}, and an R^2 of 0.9997. Compared to LSTM, M-LSTM reduces MAE and MSE by 76.3% and 58.2%, respectively, while improving R^2 by 2.3%. Compared to LSTM-Transformer, M-LSTM reduces MAE and MSE by 11.8% and 16.6%, respectively, while improving R^2 by 0.2%. The prediction performance of S-LSTM and M-LSTM is almost the same. In contrast, the LSTM model records the highest MAE and MSE values and the lowest R^2.

Fig. 11 shows the multi-step forecast performance metrics for various forecast models using 70% of the training set. Among these, M-LSTM showed the best prediction performance, with a MAE of 0.6252×10^{-3}, an MSE of 0.1858×10^{-5}, and an R^2 of 0.9987. Compared to LSTM, M-LSTM reduces MAE and MSE by 75.7% and 66.1%, respectively, while improving R^2 by 6.7%. Compared to LSTM-Transformer, M-LSTM reduces MAE and MSE by 26.6% and 39%, respectively, while improving R^2 by 1.7%. The prediction performance of S-LSTM and M-LSTM is

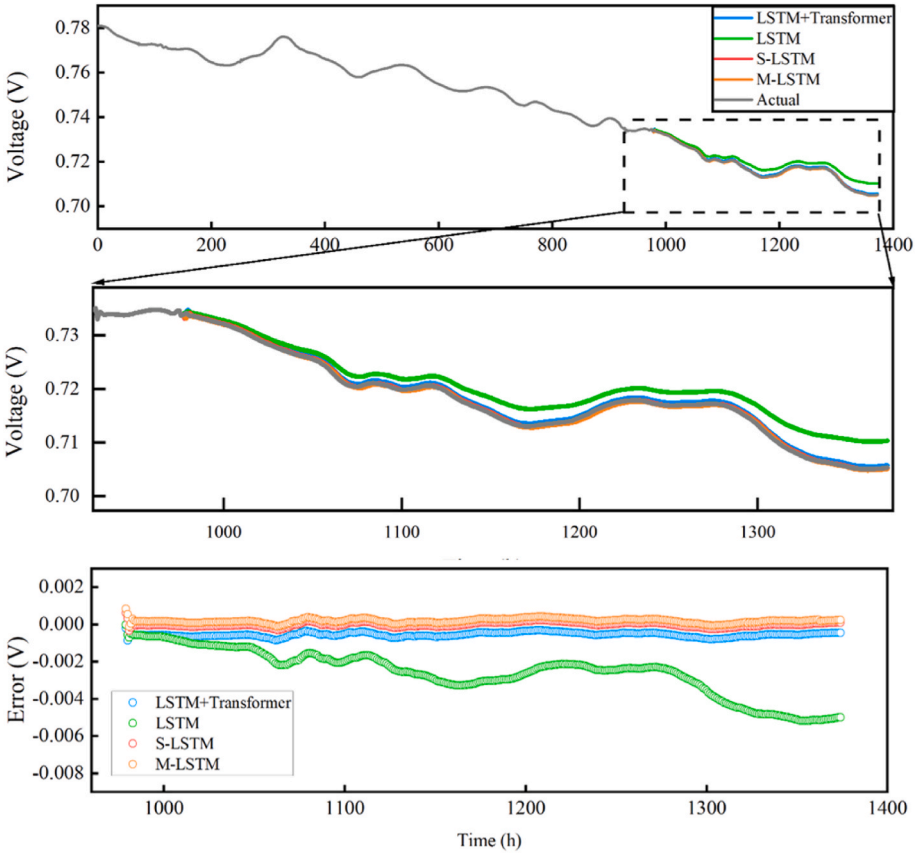


Fig. 9. 70% training set voltage decay prediction results.

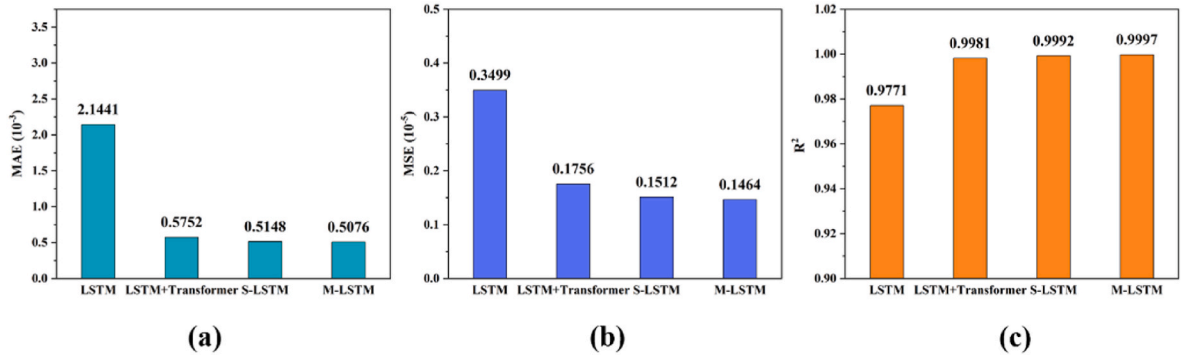


Fig. 10. One-step forecast, 70% training set:(a) MAE; (b) MSE; (c) $R.^2$.

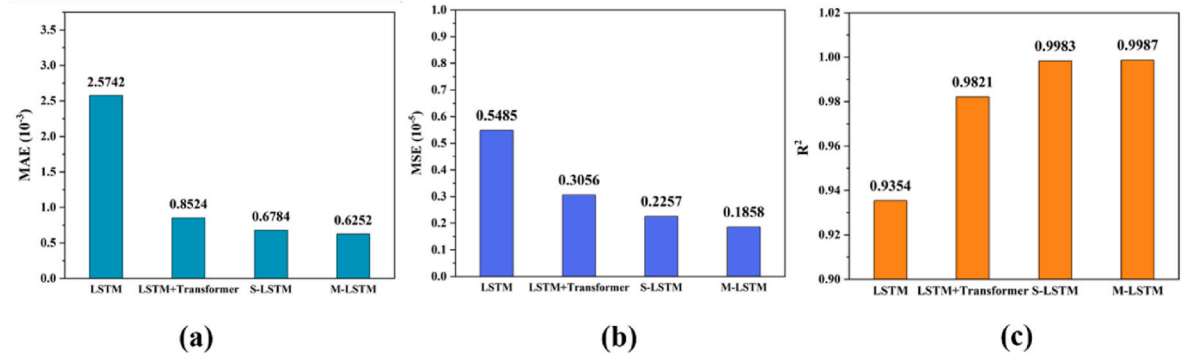


Fig. 11. Multi-step forecast, 70% training set:(a) MAE; (b) MSE; (c) $R.^2$.

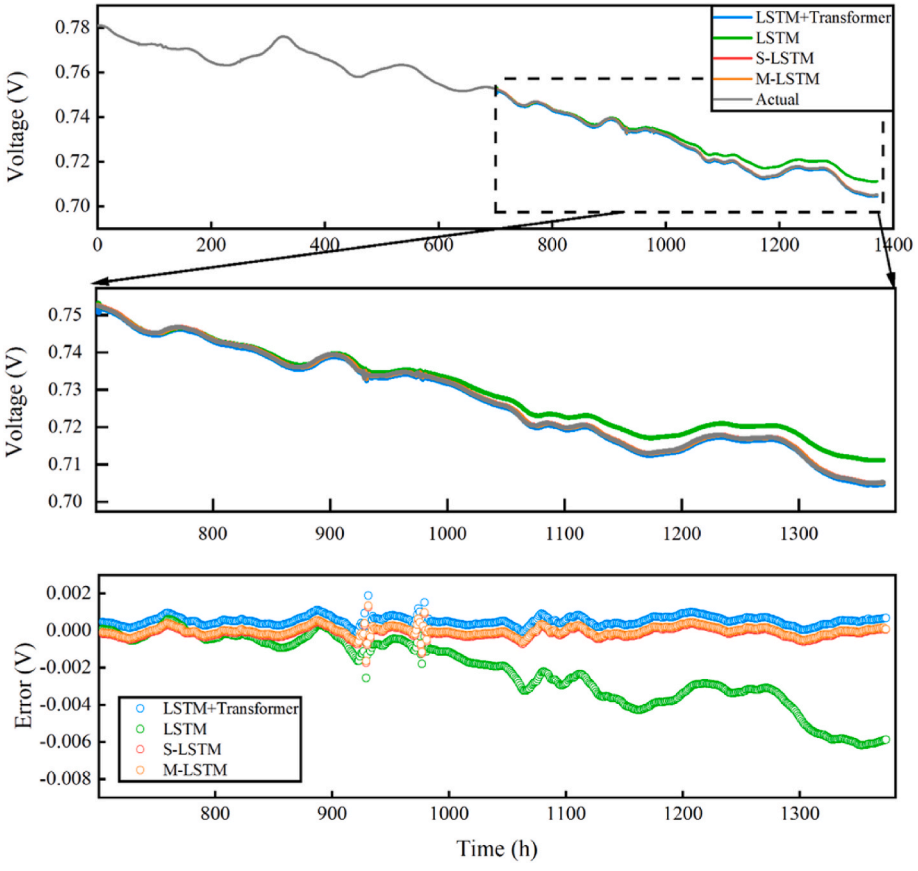


Fig. 12. 50% training set voltage decay prediction results.

almost the same. In contrast, the LSTM model records the highest MAE and MSE values and the lowest R^2.

Secondly, this paper uses 50% of the dataset as the training set and 50% as the test set. The prediction performance of LSTM, LSTM-Transformer, S-LSTM, and M-LSTM is compared. The purpose is to compare whether each model can make longer predictions when the amount of data is small. The prediction results of fuel cell degradation are shown in Fig. 12. The prediction results of S-LSTM and M-LSTM are still close to the actual monitoring voltage, and the prediction stability is higher than that of other models. In contrast, the performance of the LSTM model is the worst.

Fig. 13 shows the one-step forecast performance metrics for various forecast models using 50% of the training set. Among these, M-LSTM showed the best prediction performance, with MAE of 0.5183×10^{-3}, MSE of 0.2556×10^{-5}, and R^2 of 0.9993. Compared with LSTM, M-LSTM reduces MAE and MSE by 80.8% and 56.6%, respectively, and increases R^2 by 3.5%. Compared to LSTM-Transformer, M-LSTM reduces MAE and MSE by 21.3% and 15.5%, respectively, while improving R^2 by 0.2%. The prediction performance of S-LSTM and M-LSTM is almost the same. In contrast, the LSTM model records the highest MAE and MSE values and the lowest R^2.

Fig. 14 shows the multi-step forecast performance metrics for various forecast models using 50% of the training set. Among these, M-LSTM showed the best prediction performance, with a MAE of 0.6354×10^{-3}, an MSE of 0.3138×10^{-5}, and an R^2 of 0.9982. Compared to LSTM, M-LSTM reduces MAE and MSE by 79.4% and 60.4%, respectively, while improving R^2 by 5.2%. Compared to LSTM-Transformer, M-LSTM reduces MAE and MSE by 32% and 24.3%, respectively, while improving R^2 by 2%. The prediction performance of S-LSTM and M-LSTM is almost the same. In contrast, the LSTM model records the highest MAE and MSE values and the lowest R^2.

When comparing the results between the 50% and 70% training sets, M-LSTM maintains its high performance across both datasets. With the 70% training set, M-LSTM achieved an MAE of 0.5076×10^{-3}, MSE of 0.1464×10^{-5}, and R^2 of 0.9997. This is only a slight improvement over the 50% training set, where it achieved an MAE of 0.5183×10^{-3}, MSE of 0.2556×10^{-5}, and R^2 of 0.9993. Specifically, the decrease in MAE is just 0.0107×10^{-3}, in MSE is 0.1090×10^{-5}, and in R^2 is 0.0004. This minimal loss of accuracy with the reduced dataset highlights M-LSTM's strong generalization ability, allowing it to effectively maintain high prediction performance even with less training data.

The article further compares the performance of each model under single-step and multi-step prediction. By comparing Figs. 10 and 11, Figs. 13 and 14, it can be found that the prediction performance of all models has declined in multi-step prediction, but the results show that the prediction performance of M-LSTM is still higher than that of the other three models, and the prediction results will not have a large deviation due to multi-step prediction. On the contrary, due to the limitations of its model structure, the accuracy of LSTM has dropped significantly in multi-step prediction.

In summary, the comparison results show that M-LSTM has good prediction performance, and its prediction accuracy is higher than LSTM, LSTM-Transformer and S-LSTM. There are three main reasons: First, M-LSTM uses matrices to store key-value pairs, which is similar to the attention mechanism of Transformer and can capture richer data features. Second, the covariance update rule enables M-LSTM to automatically learn key-value relationships, avoids the design of complex feature extractors, and effectively captures long-distance dependencies. Finally, M-LSTM uses the parallelism of matrix operations to accelerate training and inference, which is particularly suitable for high-dimensional data processing. The aging data of PEMFC, including voltage, current, temperature, etc., often depends on longer time series data and has higher data dimensions. The above characteristics of M-LSTM are highly matched with the characteristics of PEMFC aging data, so better prediction results are obtained.

5. Conclusion

This paper proposes a prediction model based on matrix long short-term memory (M-LSTM) network for the problem of PEMFC remaining service life prediction, and verifies its effectiveness through experiments. The main research results are summarized as follows.

1) This paper uses the Spearman correlation coefficient to reduce the dimensionality of experimental data, identify variables that are highly correlated with PEMFC voltage decay, effectively reduce data redundancy, and improve model calculation efficiency and prediction performance.
2) The M-LSTM model effectively alleviates the performance degradation of the traditional LSTM model when processing long-term sequence data and high-dimensional data sets. Experimental results show that the M-LSTM model is superior to the LSTM and LSTM-Transformer models in terms of prediction accuracy and stability, especially when dealing with sparse labels, long-distance dependencies, and complex data patterns. Specifically, compared with the LSTM model, under single-step prediction, M-LSTM reduces MAE and MSE by 76.3% and 58.2%, respectively, while increasing R^2 by 2.3%. Under multi-step prediction, M-LSTM reduces MAE and MSE by 79.4% and 60.4% respectively, while improving R^2 by 5.2%.

Compared with traditional LSTM, M-LSTM introduces additional computational overhead due to the matrix memory mechanism. In PEMFC life prediction with high real-time requirements, especially when rapid decision-making or dynamic adjustment is required, the computational complexity of M-LSTM may become a bottleneck. In the future, the model architecture can be further optimized, such as compressing the M-LSTM model or adopting lightweight deep learning methods to reduce the demand for computing resources and improve the efficiency of real-time prediction.

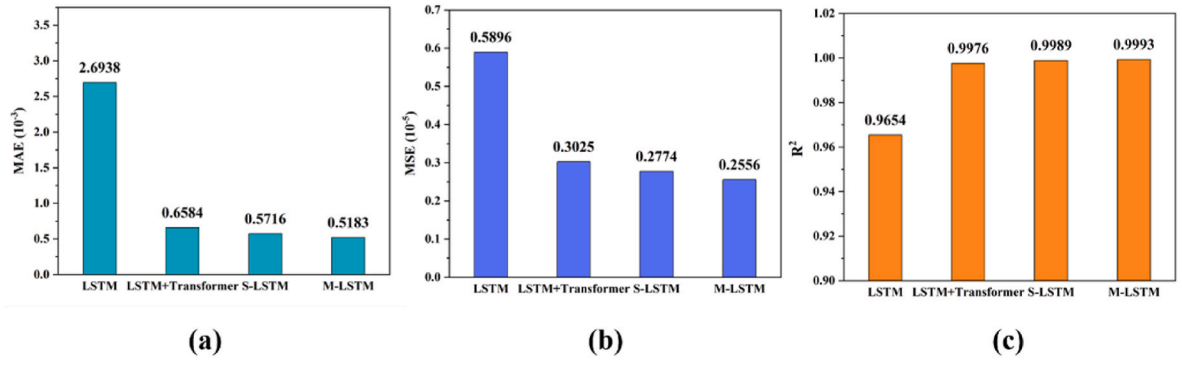


Fig. 13. One-step forecast, 50% training set:(a) MAE; (b) MSE; (c) R.^{2}.

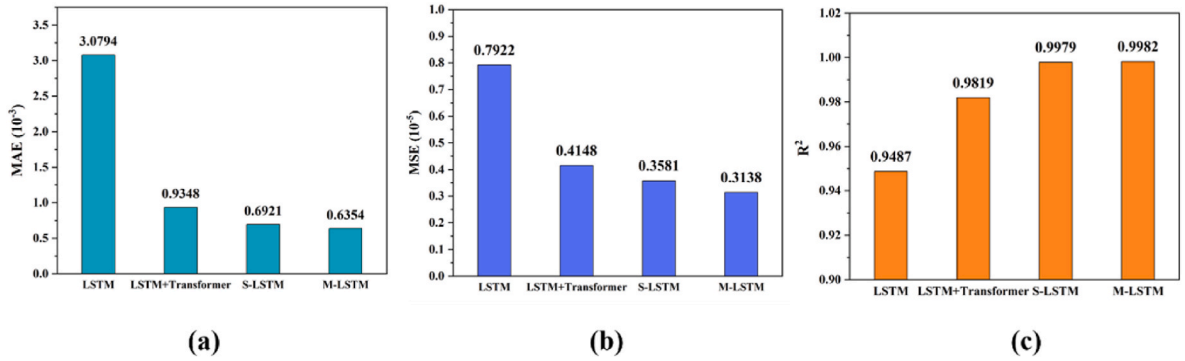


Fig. 14. One-step forecast, 50% training set:(a) MAE; (b) MSE; (c) R.^{2}.

CRediT authorship contribution statement

Fengyan Yi: Writing – original draft, Visualization, Methodology, Investigation, Funding acquisition, Formal analysis. **Xing Shu:** Visualization, Supervision, Formal analysis. **Jiaming Zhou:** Writing – review & editing, Validation, Resources, Funding acquisition, Conceptualization. **Jinming Zhang:** Writing – review & editing, Validation, Supervision. **Chunxiao Feng:** Writing – review & editing, Supervision, Resources, Conceptualization. **Hongtao Gong:** Visualization, Formal analysis. **Caizhi Zhang:** Visualization, Validation. **Wenhao Yu:** Software, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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